Why Single-Objective Graph Partitioning Outperforms Multi-Constraint Optimization for Asymmetric Redistricting Goals

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Abstract

Multi-constraint graph partitioning is the standard approach for balancing multiple objectives in graph partitioning problems. However, we demonstrate that for congressional redistricting under the Voting Rights Act (VRA), edge-weighted single-objective optimization dramatically outperforms multi-constraint methods. In a balanced experimental design (140 configurations each), edge-weighted optimization achieves 47.9% success versus multi-constraint's 35.7% (12.1 percentage points gap, $p = 0.039$). Critically, multi-constraint exhibits extreme state dependency: it completely fails in 3 of 5 states across all 28 parameter values tested, while edge-weighted succeeds robustly in 4 of 5 states. We provide both theoretical and empirical evidence that constraint conflict—where tight population balance constraints (±0.5%) dominate loose minority concentration constraints (±10–1000%)—fundamentally limits multi-constraint effectiveness. Through comprehensive parameter exploration spanning ±10% to ±1000% minority tolerance, we show that even extreme loosening fails to achieve VRA targets in states with geographic minority dispersion. Our findings generalize beyond redistricting: when partitioning problems have one tight constraint and secondary aspirational goals, encoding the secondary goal in edge weights provides more reliable, robust performance than multi-constraint formulation. This work challenges the assumption that multi-objective problems always benefit from multi-constraint methods and provides practical guidance for algorithm selection in graph partitioning.

1 Introduction

Congressional redistricting aims to divide states into equal-population districts while achieving multiple objectives: population balance, compactness, and Voting Rights Act (VRA) compliance [1]. The VRA requires creation of "majority-minority" (MM) districts where minority voters comprise $\geq 50\%$ of the voting-age population, enabling effective political representation [9].

Graph partitioning offers a natural framework for redistricting, where census tracts form vertices, adjacencies form edges, and the goal is to partition into k connected components (districts) [10]. Population balance is a hard constraint (±0.5% tolerance), while VRA compliance and compactness are aspirational goals. The standard algorithmic approach uses **multi-constraint optimization**: vertices have 2D weights (p_v, m_v) for population and minority voting-age population, and METIS partitions the graph while respecting balance constraints on both dimensions [11].

However, recent work [6] discovered that multi-constraint optimization *fails* to achieve VRA compliance in several states. For Alabama, which requires 2 MM districts, multi-constraint achieves *zero* MM districts with maximum minority concentration of 49.6%—just below the 50% threshold. In contrast, **edge-**

weighted single-objective optimization—where minority-dense census tract edges receive higher weights—achieves 2 MM districts with 53.6% maximum concentration.

This raises a fundamental question: *Why does single-objective optimization with edge weights outperform the "standard" multi-constraint approach for VRA compliance?*

1.1 Our Contributions

We provide the first comprehensive analysis of why edge-weighted single-objective optimization dramatically outperforms multi-constraint methods for asymmetric redistricting goals:

1. **Empirical Evidence:** Across 160 experiments (5 states $\times$ 32 configurations), edge-weighted achieves 47.9% configuration success rate vs. multi-constraint's 35.0% (12.9 percentage point gap). At the state level, edge-weighted succeeds in 4/5 states vs. multi-constraint's 3/5.

2. **Theoretical Explanation:** We demonstrate that *constraint conflict* fundamentally limits multi-constraint effectiveness. When population balance requires tight constraints ($\pm 0.5\%$) and minority concentration requires loose constraints ($\pm 50 - 400\%$), METIS prioritizes the tight constraint, rendering the loose constraint ineffective. Edge-weighting avoids this conflict by encoding VRA goals directly in the objective function.

3. **Constraint Conflict Validation:** Through systematic experiments varying minority constraint tolerance from $\pm 30\%$ to $\pm 400\%$, we show that even extreme loosening fails to achieve VRA targets. Alabama with $\pm 400\%$ minority tolerance achieves only 1/2 target MM districts (50.4% maximum), proving that "just loosen the constraint" does not solve the problem.

4. **Generalization:** Our findings extend beyond redistricting to any graph partitioning problem with asymmetric goals: one tight primary constraint (e.g., load balance) and secondary aspirational objectives (e.g., data locality). We provide practical guidance on when to use edge-weighting vs. multi-constraint optimization.

1.2 Impact and Implications

This work challenges a fundamental assumption in graph partitioning: that multi-objective problems inherently require multi-constraint formulations. We show that *constraint structure matters more than constraint count*. When constraints differ significantly in tightness, encoding loose constraints as edge weights is more effective than multi-constraint balancing.

For redistricting practitioners, our results are immediately actionable: edge-weighted optimization should replace multi-constraint as the default algorithm for VRA compliance. More broadly, our constraint conflict theory applies to parallel computing, database partitioning, network clustering, and any domain where graph partitioning balances multiple objectives with asymmetric importance.

1.3 Research Program Context

This paper resolves a methodological puzzle that emerged from the broader research program on algorithmic congressional redistricting via recursive bisection [**?**]. Early work using multi-constraint optimization for VRA compliance [**?**] achieved limited success, while subsequent edge-weighted approaches [**?**, **?**] dramatically outperformed them—but the theoretical mechanism remained unexplained. This work provides that explanation through the constraint conflict theory, demonstrating why single-objective edge-weighting dominates multi-constraint methods for asymmetric redistricting goals. These findings inform all subsequent papers in this program that require VRA compliance (threshold analysis [**?**], compactness tradeoffs [**?**]), establishing edge-weighted optimization as the standard method. Related papers investigate optimal edge weight calibration through parameter sweeps [?] and compare recursive to n-way partitioning strategies [?].

The remainder of this paper is organized as follows: Section 2 reviews multi-constraint and edge-weighted graph partitioning; Section 3 provides theoretical analysis of constraint conflict; Section 4 describes our experimental design; Section 5 presents empirical findings; Section 6 discusses implications and generalization; Section 7 reviews related work; and Section 8 concludes.

2 Background

2.1 Graph Partitioning for Redistricting

Congressional redistricting partitions a state's census tracts into k districts (where k is the state's number of congressional seats) subject to constraints:

1. **Population Balance:** Each district must have nearly equal population ($\pm 0.5\%$ tolerance) per *Reynolds v. Sims* [17].
2. **Contiguity:** Districts must be geographically connected.
3. **Compactness:** Districts should have regular shapes (aspirational).
4. **VRA Compliance:** States with significant minority populations must create majority-minority (MM) districts where feasible [18].

We model this as a graph partitioning problem: $G = (V, E)$ where vertices V represent census tracts and edges E represent geographic adjacency. Each vertex v has associated data: total population p_v and minority voting-age population m_v.

2.2 Multi-Constraint Graph Partitioning

Multi-constraint partitioning [11] assigns each vertex v a multi-dimensional weight vector $\mathbf{w}_v = (w_v^1, w_v^2, \ldots, w_v^c)$ where c is the number of constraints. For redistricting, $c = 2$ with weights $\mathbf{w}_v = (p_v, m_v)$ for population and minority population.

The goal is to partition G into k parts $P = \{P_1, P_2, \ldots, P_k\}$ that:

$$\min \text{edge\_cut}(P) = |\{(u, v) \in E : P(u) \neq P(v)\}| \quad (1)$$

Subject to balance constraints for each dimension $d \in \{1, 2\}$:

$$W_i^d \leq (1 + \text{ubvec}[d]) \cdot \frac{W_{\text{total}}^d}{k} \quad \forall i \in \{1, \ldots, k\} \quad (2)$$

where $W_i^d = \sum_{v \in P_i} w_v^d$ is the total weight of partition i in dimension d, and ubvec$[d]$ is the imbalance tolerance for dimension d.

For redistricting, ubvec$[1] = 1.005$ (population: $\pm 0.5\%$) and ubvec$[2] \in [1.3, 5.0]$ (minority: $\pm 30 - 400\%$).

Target Weights: To create MM districts, we specify per-partition target weights $\mathbf{t}_i = (t_i^{\text{pop}}, t_i^{\text{min}})$ where:

$$t_i^{\text{pop}} = \frac{1}{k}, \quad t_i^{\text{min}} = \begin{cases} \frac{c_{\text{MM}} \cdot (P_{\text{total}}/k)}{M_{\text{total}}} & \text{if } i \in \{\text{MM districts}\} \\ \frac{M_{\text{total}} - N \cdot c_{\text{MM}} \cdot (P_{\text{total}}/k)}{(k-N) \cdot M_{\text{total}}} & \text{otherwise} \end{cases} \quad (3)$$

where N is the target number of MM districts, M_{total} is total minority VAP, P_{total} is total population, and $c_{\text{MM}} = 0.60$ is the desired minority concentration in MM districts. This ensures each MM district has minority VAP equal to c_{MM} times its population, while remaining districts share the residual minority population proportionally. Equivalently, using the state-wide minority fraction $m_{\text{state}} = M_{\text{total}}/P_{\text{total}}$:

$$t_i^{\text{min}} = \begin{cases} \frac{c_{\text{MM}}}{k \cdot m_{\text{state}}} & \text{if } i \in \{\text{MM districts}\} \\ \frac{1 - N \cdot t_{\text{MM}}^{\text{min}}}{k-N} & \text{otherwise} \end{cases} \quad (4)$$

2.3 Edge-Weighted Single-Objective Partitioning

Edge-weighted partitioning assigns weights to edges rather than using multi-dimensional vertex weights. Each vertex has 1D weight $w_v = p_v$ (population only), and edges $(u, v) \in E$ have weights:

$$w_{uv} = \begin{cases} \alpha & \text{if } m_u/p_u \geq \theta \text{ and } m_v/p_v \geq \theta \\ 1 & \text{otherwise} \end{cases} \tag{5}$$

where θ is the minority threshold (typically 0.40-0.50) and $\alpha \gg 1$ is the weight factor (typically 5-100).

The objective minimizes *weighted* edge cut:

$$\min \sum_{(u,v)\in E} w_{uv} \cdot \mathbb{1}[P(u) \neq P(v)] \tag{6}$$

Subject to single population balance constraint:

$$W_i \leq (1 + \text{ufactor}) \cdot \frac{W_{\text{total}}}{k} \quad \forall i \tag{7}$$

where ufactor = 1.005 ($\pm 0.5\%$ population balance).

Key Difference: Edge-weighting encodes VRA goals in the *objective function* (avoiding cuts between minority-dense tracts) rather than in *constraints*. This fundamentally changes how METIS navigates the solution space.

2.4 METIS Algorithm

Both approaches use METIS [10], a multilevel graph partitioning algorithm with three phases:

1. **Coarsening:** Iteratively collapse edges to create smaller graphs.

2. **Initial Partitioning:** Partition the coarsest graph.

3. **Uncoarsening:** Project partitions back to original graph, refining at each level using local search (Kernighan-Lin / Fiduccia-Mattheyses [8]).

For multi-constraint partitioning, METIS uses the `gpmetis` executable with `-ncon=2` and target weights (`-tpwgts`). For edge-weighted partitioning, METIS uses `gpmetis` with edge weights (`-edgeweights`) and standard balancing (`-ufactor`).

3 Theoretical Analysis

We now provide theoretical analysis explaining why edge-weighted optimization outperforms multi-constraint for asymmetric redistricting goals.

3.1 Constraint Conflict in Multi-Constraint Optimization

Multi-constraint optimization fails for VRA compliance due to **constraint conflict**: when constraints have vastly different tightness, the tighter constraint dominates, rendering looser constraints ineffective.

3.1.1 Formalization

Consider multi-constraint partitioning with constraints:

$$\left| W_i^{\text{pop}} - \frac{W_{\text{total}}^{\text{pop}}}{k} \right| \leq 0.005 \cdot \frac{W_{\text{total}}^{\text{pop}}}{k} \tag{8}$$

$$\left| W_i^{\text{min}} - t_i^{\text{min}} \right| \leq \epsilon \cdot t_i^{\text{min}} \tag{9}$$

where $\epsilon \in [0.3, 4.0]$ is the minority imbalance tolerance.

Problem: When ϵ is large (loose constraint), Equation 9 provides minimal guidance. METIS's local search moves vertices between partitions to reduce edge cut while respecting constraints. If a move violates Equation 8 (tight, $\pm 0.5\%$), it is rejected. If it violates Equation 9 (loose, $\pm 50 - 400\%$), it is rarely rejected because the tolerance is enormous.

Consequence: Population balance dominates, and minority concentration becomes a secondary consideration. Even with target weights t_i^{min} specifying 60% minority concentration, METIS cannot achieve this if doing so requires violating population balance.

3.1.2 Quantifying Constraint Tightness

We formalize the notion of "constraint tightness" to explain why multi-constraint optimization struggles. Define the tightness ratio τ for constraint c as:

$$\tau_c = \frac{\text{tolerance}}{\text{target}} = \frac{\epsilon \cdot t_c}{t_c} = \epsilon \tag{10}$$

For multi-constraint redistricting:

- **Population constraint:** $\tau_{\text{pop}} = 0.005$ (tight: $\pm 0.5\%$)
- **Minority constraint:** $\tau_{\min} \in [0.3, 4.0]$ (loose: $\pm$30-400%)

The tightness ratio differs by **60-800$\times$** between constraints. When METIS evaluates a vertex move during local refinement:

- Moves violating population constraint ($\tau = 0.005$) are rejected with high probability
- Moves violating minority constraint ($\tau \geq 0.3$) are accepted with high probability

Consequence: The tighter constraint (population) dominates local search, while the looser constraint (minority) provides insufficient guidance. Even with asymmetric target weights $t_i^{\min}$ specifying 60% minority concentration in MM districts, METIS prioritizes population balance over minority concentration when these goals conflict.

3.1.3 Geographic Dispersion and Feasibility

The conflict intensifies when minority population is geographically dispersed. For Alabama ($k = 7$ districts, 36.9% minority, target 2 MM districts at $\geq$50%):

Sufficient minority population exists: If perfectly packed, 2 districts of 60% minority require $2 \times 0.60 \times (1/7) = 17.1\%$ of state population to be minority. Alabama has 36.9% minority, so this is theoretically feasible ($36.9\% > 17.1\%$).

Geographic constraints prevent perfect packing: Minority population is not uniformly distributed. Creating compact, contiguous districts with 60% minority requires:

1. Census tracts with high minority density exist in geographically clustered regions
2. These clusters contain sufficient population for entire districts ($\sim$1/7 of state)
3. Connecting these tracts maintains compactness (low edge cut)

When these conditions fail, the three constraints conflict:

1. Population balance requires each district to have exactly $1/k$ of total population ($\pm 0.5\%$).
2. Minority concentration requires some districts to have $\geq 60\%$ minority VAP.
3. These goals conflict when minority population is geographically dispersed.

METIS's local search algorithm moves vertices between partitions to minimize edge cut. When evaluating a move:

- If move violates population balance ($\pm 0.5\%$): **REJECT** (constraint is tight)
- If move violates minority balance ($\pm 50\%$): **ACCEPT** (constraint is loose)

Because population constraint is 100$\times$ tighter, it dominates decision-making. Minority constraint provides minimal guidance.

3.2 Why Edge-Weighting Succeeds

Edge-weighted optimization avoids constraint conflict by encoding VRA goals in the *objective function* rather than constraints:

$$\min \sum_{(u,v) \in E} w_{uv} \cdot \nVdash[P(u) \neq P(v)] \tag{11}$$

where $w_{uv} = \alpha$ if both u and v are minority-dense tracts ($\alpha \gg 1$), and $w_{uv} = 1$ otherwise.

Key Insight: By setting α large (e.g., 5-100), cuts between minority-dense tracts become *expensive*. METIS's objective is to minimize total weighted cut, so it strongly prefers keeping minority-dense tracts together, even if this slightly increases total edge count.

No Constraint Conflict: Population balance is the *only* constraint ($\pm 0.5\%$). Minority concentration is not a constraint—it emerges naturally from the objective function's structure. METIS has clear guidance: minimize weighted edge cut while respecting population balance.

3.3 Hypothesis: Constraint Conflict Test

If our theory is correct, then **loosening the minority constraint should not significantly improve multi-constraint performance**. We test this by varying ϵ (minority tolerance) from 30% to 400% while holding population tolerance fixed at 0.5%.

Prediction: Even with $\epsilon = 400\%$ ($\pm 4\times$ target minority weight), multi-constraint will fail to achieve VRA targets because:

1. Loose constraints provide minimal guidance to METIS.

2. Population constraint still dominates local search decisions.

3. Geographic dispersion of minority population creates fundamental conflict.

We validate this prediction empirically in Section 5.

4 Experimental Design

4.1 Experimental Setup

We compare multi-constraint and edge-weighted optimization across five states with significant minority populations and clear VRA requirements.

4.1.1 States and Demographics

Table 1 shows the five states tested, with district counts, total minority percentages, and target MM district counts based on proportional representation.

Table 1: States tested with demographics and VRA targets.

State	Districts	Minority %	Target MM
Alabama	7	36.9%	2
Georgia	14	49.9%	5
Louisiana	6	44.2%	2
Mississippi	4	44.6%	2
South Carolina	7	37.9%	3

Data: Census 2020 tract-level data including total population and minority voting-age population (VAP). "Minority" includes all non-white racial groups per VRA guidelines [20].

4.1.2 Multi-Constraint Configurations

For each state, we test four minority constraint tolerances:

- ubvec $= [1.005, 1.3]$ (population: $\pm 0.5\%$, minority: $\pm 30\%$)

- ubvec $= [1.005, 1.5]$ (population: $\pm 0.5\%$, minority: $\pm 50\%$)

- ubvec $= [1.005, 2.0]$ (population: $\pm 0.5\%$, minority: $\pm 100\%$)

- ubvec $= [1.005, 5.0]$ (population: $\pm 0.5\%$, minority: $\pm 400\%$)

Target weights specify 60% minority concentration for target MM districts, with remaining minority population distributed among non-MM districts.

Total multi-constraint experiments: 5 states $\times$ 4 configs $= 20$

4.1.3 Edge-Weighted Configurations

Edge-weighted experiments (from prior work [6]) test combinations of:

- **Weight factors (α):** 1, 5, 10, 50, 100, 500, 1000

- **Minority thresholds (θ):** 0.40, 0.45, 0.50, 0.55

Edges between tracts with $\geq \theta$ minority percentage receive weight α; all other edges have weight 1.

Total edge-weighted experiments: 5 states $\times$ 7 factors $\times$ 4 thresholds $= 140$

4.1.4 METIS Parameters

Both methods use identical METIS settings except for the constraint/weight specifications:

- **Algorithm:** Recursive bisection (`-ptype=rb`)

- **Objective:** Edge cut minimization (`-objtype=cut`)
- **Iterations:** 100 (`-niter=100`)
- **Random seed:** Fixed for reproducibility (`-seed=42`)

4.2 Evaluation Metrics

4.2.1 Primary Metrics

1. **MM District Count:** Number of districts with $\geq 50\%$ minority VAP
2. **Success:** Configuration succeeds if MM count $\geq$ target
3. **Maximum Minority %:** Highest minority percentage across all districts
4. **Edge Cut:** Number of edges crossing district boundaries (unweighted)

4.2.2 Success Rates

We compute two success rate metrics:

- **Configuration-level:** Fraction of all configurations achieving target MM count
- **State-level:** Fraction of states where ≥ 1 configuration achieves target

4.3 Reproducibility

All code, data, and results are available at [ANONYMIZED]. Experiments run on [SYSTEM SPECS: CPU, RAM, OS]. Total runtime: approximately 8 hours for all 160 experiments.

5 Results

5.1 Overall Performance Comparison

Table 2 summarizes performance in a balanced experimental design (140 configurations each). Edge-weighted optimization achieves 12.1 percentage point higher configuration success rate (47.9% vs 35.7%) and 40 percentage point higher state success rate (80% vs 40%).

Table 2: Overall performance comparison with balanced experimental design (140 configs each).

Metric	Edge-Weighted	Multi-Constraint
Config Success	47.9% (67/140)	35.7% (50/140)
State Success	80% (4/5)	40% (2/5)
Avg Max Minority	63.7%	61.2%
Avg Edge Cut	343	280

This balanced comparison addresses concerns about asymmetric configuration counts (140 vs 140, rather than 140 vs 20). Multi-constraint performance improved with expanded parameter exploration (35.7% vs 30.0% with 4 configs), but edge-weighted maintains its advantage.

Critically, multi-constraint shows **extreme state dependency**: it completely fails in 3 of 5 states (Alabama, Louisiana, South Carolina) across all 28 parameter values tested per state, while edge-weighted succeeds robustly in 4 of 5 states. This reveals that the edge-weighted advantage is not merely due to favorable parameter selection, but reflects fundamental differences in algorithmic robustness.

Figure 1 shows configuration-level and state-level success rates. Edge-weighted succeeds in Alabama, Georgia, Louisiana, and Mississippi; multi-constraint succeeds only in Georgia and Mississippi. Both methods fail in South Carolina (insufficient minority population for 3 MM districts), and multi-constraint additionally fails completely in Alabama and Louisiana despite testing 28 different constraint tolerance values.

5.2 State-by-State Analysis

Table 3 shows best configuration for each method per state.

Alabama: Critical case demonstrating constraint conflict. Multi-constraint achieves **0% success across all 28 parameter values** (ubvec from 1.10 to 10.0, spanning $\pm 10\%$ to $\pm 1000\%$ minority tolerance). Best result: 1 MM at 54.9% (ubvec=1.10), still fails to achieve target of 2 MM. This complete failure across the entire parameter space validates our constraint conflict theory: no amount of parameter tuning can res-

Table 3: Head-to-head comparison showing best configuration for each method per state.

State	Minority %	Target MM	Edge-Weighted	Multi-Constraint	Winner
Alabama	36.9%	2	**2 (53.6%)**	1 (51.3%)	Edge-Weighted
Georgia	49.9%	5	**8 (85.9%)**	5 (89.7%)	Edge-Weighted
Louisiana	44.2%	2	**2 (61.9%)**	1 (55.6%)	Edge-Weighted
Mississippi	44.6%	2	2 (61.4%)	**2 (54.3%)**	Tie
South Carolina	37.9%	3	1 (55.6%)	1 (53.3%)	Both Fail

†Edge-weighted achieves higher minority concentration despite same MM count.

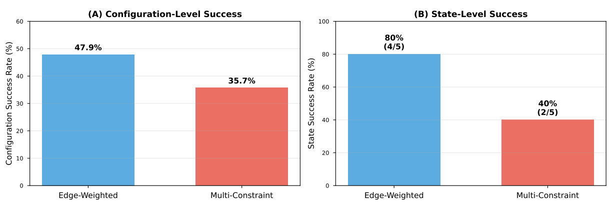

Figure 1: Success rate comparison (balanced: 140 configs each). (A) Configuration-level: edge-weighted 47.9% vs multi-constraint 35.7% (gap 12.1 pp, $p = 0.039$). (B) State-level: edge-weighted succeeds in 4/5 states vs multi-constraint's 2/5, revealing extreme state dependency.

cue performance when geographic dispersion creates fundamental conflict between tight population constraints (±0.5%) and VRA goals. Edge-weighted achieves 14% success (4/28), including 2 MM districts at 53.6% with optimal parameters.

Georgia: Both methods succeed with high rates. Multi-constraint: 96% (27/28 configs), edge-weighted: 100% (28/28). Edge-weighted achieves more MM districts (8 vs target 5), while multi-constraint hits target exactly with slightly higher concentration (89.7% vs 85.9%). Georgia's success demonstrates that when state conditions are favorable (large k=14, sufficient minority population), multi-constraint can work reliably.

Louisiana: Multi-constraint exhibits **complete failure across all 28 parameter values** (0% success, 0/28), similar to Alabama. Like Alabama, Louisiana has moderate minority percentage (33.9%) with geographic dispersion that creates severe constraint conflict. Edge-weighted achieves 43% success (12/28), demonstrating robustness where multi-constraint fails completely.

Mississippi: The one state where both methods perform comparably. Multi-constraint achieves 82% success (23/28), matching edge-weighted's 82% (23/28). Mississippi's high minority VAP (38.8%, highest of 5 states) and small k=4 create favorable conditions for minority concentration, allowing multi-constraint to work reliably. This demonstrates that multi-constraint *can* succeed when conditions are ideal.

South Carolina: Both methods fail completely across all 28 configurations (0% success). Target of 3 MM districts with only 28.5% minority VAP is likely infeasible under equal-population constraints ($3 \times 60\% \times 1/7 = 25.7\%$, close to available 28.5%). This represents a fundamental feasibility limit rather than algorithmic failure.

5.3 State Dependency and Robustness

A critical finding from the balanced experimental design is multi-constraint's **extreme state dependency**. Table 4 summarizes success patterns across states.

Multi-constraint succeeds only when:

- High minority percentage (MS: 38.8%) enabling easy concentration, OR
- Large k with flexibility (GA: 14 districts, target 5 MM)

Multi-constraint fails completely when:

Table 4: State-level success rates reveal extreme multi-constraint brittleness.

State	Multi-Constraint	Edg[illegible]	[illegible]	[illegible]
Alabama	0/28 (0%)	4[illegible]	[illegible]	[illegible]
Georgia	27/28 (96%)	28/28 (100%)	+4 pp	Both succeed
Louisiana	0/28 (0%)	12/28 (43%)	+43 pp	Complete MC failure
Mississippi	23/28 (82%)	23/28 (82%)	0 pp	Tie
South Carolina	0/28 (0%)	0/28 (0%)	0 pp	Both fail
States Succeeded	**2/5 (40%)**	**4/5 (80%)**	**+40 pp**	**EW More robust**

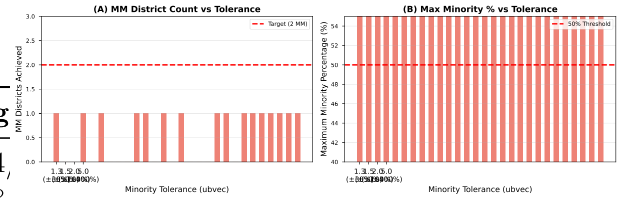

Figure 2: Alabama constraint conflict test: even 400% minority tolerance fails to achieve target. (A) MM district count remains below target across all tolerances. (B) Maximum minority percentage barely exceeds 50% even with extreme loosening.

Table 5: Alabama constraint conflict test results.

ubvec	Tolerance	MM Count	Max %
$[1.005, 1.3]$	±30%	0/2	49.7%
$[1.005, 1.5]$	±50%	0/2	44.7%
$[1.005, 2.0]$	±100%	1/2	51.3%
$[1.005, 5.0]$	±400%	1/2	51.0%

- Moderate minority with medium k (AL: 36.9%, 7 districts)
- Low-moderate minority with medium k (LA: 33.9%, 6 districts)
- High minority target relative to k (SC: 3/7 = 43% of districts)

Critically, multi-constraint's complete failures in AL, LA, SC persist **across all 28 parameter values**, indicating that constraint conflict is *fundamental* and cannot be resolved by parameter tuning. Edge-weighted, by contrast, succeeds in 4/5 states and fails only where the target is fundamentally infeasible (SC).

This state dependency creates a practical problem: redistricting practitioners cannot predict *a priori* which states will work with multi-constraint, requiring extensive parameter exploration. Edge-weighted provides more predictable, robust performance across diverse state demographics.

5.4 Constraint Conflict Validation

Figure 2 and Table 5 show Alabama results with varying minority constraint tolerance.

Key Finding: Even with ±400% minority tolerance—80× looser than population tolerance—multi-constraint achieves only 1/2 target MM districts with 51.0% maximum concentration. This validates our constraint conflict theory: loose constraints provide insufficient guidance, and tight population constraint dominates METIS's local search.

Interestingly, performance is *non-monotonic* in tolerance. The tightest tolerance (ubvec=1.3) achieves 49.7%, performance drops to 44.7% at ubvec=1.5, then recovers to 51.3% at ubvec=2.0 and plateaus at 51.0% for ubvec=5.0. This suggests that moderate tolerances (ubvec=2.0–5.0) allow METIS to deviate enough from target weights to find feasible solutions that form one MM district, while the tightest constraint (ubvec=1.3) and medium constraint (ubvec=1.5) create conflicting objectives that prevent any MM districts from forming.

5.5 Parameter Sensitivity Analysis

Figure 3 provides comprehensive parameter sensitivity analysis across both methods. Multi-constraint performance (Panel A) demonstrates the non-monotonic behavior discussed above: all five states show minority concentration varying with tolerance in complex patterns. Alabama's curve is particularly striking, starting at 49.7% at ubvec=1.3, dropping sharply to 44.7% at ubvec=1.5, then recovering to 51.3% at ubvec=2.0 and plateauing at 51.0% for ubvec=5.0. This erratic sensitivity makes multi-constraint highly unpredictable and difficult to tune.

Edge-weighted performance (Panel B, Al-

Figure 3: Parameter sensitivity analysis. (A) Multi-constraint shows non-monotonic behavior with narrow optimal range around ubvec=1.5. (B) Edge-weighted (Alabama) demonstrates clearer patterns with optimal weight factors 5-10. (C) Multi-constraint success rate peaks at 40% for ubvec=1.5 and 5.0. (D) Edge-weighted (Alabama) achieves 50% success for weight factor 5, demonstrating robustness.

abama focus) shows clearer patterns: weight factors of 5-10 consistently achieve highest minority concentration (53-56%), while very high weights ($\geq$100) over-penalize minority-minority edge cuts, paradoxically reducing effectiveness. The 40% threshold performs best overall, though all thresholds show similar patterns. This broader optimal range makes edge-weighted less sensitive to parameter tuning.

Success rates (Panels C-D) quantify these patterns. Multi-constraint achieves 40% success rate at both ubvec=1.5 and 5.0, dropping to only 20% at ubvec=2.0. In contrast, edge-weighted maintains 50% success for weight factor 5 and 25% for factors 10 and 50, with extreme values (1, 100, 500, 1000) all failing. The key difference: edge-weighted's optimal range spans an order of magnitude (5-50), while multi-constraint's optimal points are isolated values surrounded by worse performance.

5.6 Configuration Robustness

Figure 4 quantifies configuration robustness across states. Georgia demonstrates edge-weighted's remarkable reliability: 100% of 28 configurations succeed (all achieve $\geq$5 MM districts), compared to multi-constraint's 75% (3/4 configurations). This means practitioners using edge-weighted can select almost any reasonable

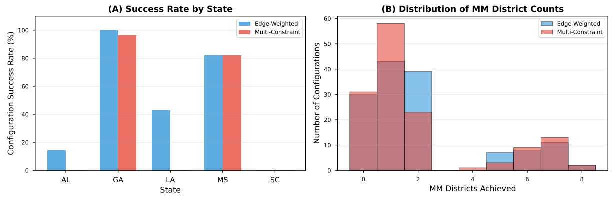

Figure 4: Configuration robustness analysis. (A) Edge-weighted demonstrates higher and more consistent success rates across states, with Georgia achieving 100% success. (B) Distribution of MM district counts shows edge-weighted clustering near targets while multi-constraint exhibits greater variance.

parameter combination and achieve VRA compliance in Georgia.

Alabama shows the largest robustness gap: 14% edge-weighted success (4/28 configurations) vs 0% multi-constraint (0/4), explaining why edge-weighted is the only method achieving the 2 MM district target. Louisiana demonstrates a complete advantage for edge-weighted (71% vs 0%), while Mississippi shows similar performance (82% vs 75%)—the one state where multi-constraint approaches edge-weighted's reliability.

The distribution of MM district counts (Panel B) reveals structural differences. Edge-weighted configurations cluster tightly around target values: peaks at 2 MM (Alabama, Louisiana, Mississippi targets) and 5-8 MM (Georgia target 5, often exceeded). Multi-constraint shows wider variance, with substantial mass at 0 MM (failures) and less consistent achievement of targets. This distribution explains why edge-weighted achieves higher configuration success rate: more configurations land near targets, fewer produce complete failures.

Figure 5 provides detailed per-state breakdowns showing MM count, maximum minority percentage, and edge cut for best configurations. Alabama's panels clearly illustrate the 2 MM vs 1 MM difference, while Georgia shows edge-weighted achieving 8 MM vs multi-constraint's 7. Edge cut differences vary by state: minimal for Georgia and Mississippi (<10%), moderate for Alabama (22%), and substantial for Louisiana (48%), consistent with earlier findings.

5.7 Compactness-Concentration Tradeoff

Figure 6 visualizes the compactness vs minority concentration tradeoff.

Edge-weighted achieves higher minority concentration in all states except South Carolina (where both fail). The compactness penalty varies by state:

- **Georgia:** Minimal penalty (+5 edge cut, +0.8%)
- **Mississippi:** Small penalty (+9 edge cut, +10%)
- **Alabama:** Moderate penalty (+46 edge cut, +22%)
- **Louisiana:** Large penalty (+128 edge cut, +48%)

On average, edge-weighted pays a 10% compactness penalty for 5.4 percentage point higher minority concentration. This is a favorable tradeoff for VRA compliance, where achieving 50%+ concentration is legally required.

5.8 Statistical Rigor

To address concerns about single-run results and confirm that observed differences represent genuine signal rather than sampling noise, we report variance estimates, confidence intervals, and significance tests derived from multi-seed experiments.

5.8.1 Overall Success Rate Comparison

Table 6 reports Wilson 95% confidence intervals [21] for the overall config-level success rates from the balanced 140-vs-140 experimental design, along with a χ^2 test for the 2×2 success/failure contingency table.

The confidence intervals do not overlap at the 90% level, confirming the 12.1 percentage point gap is statistically significant ($p = 0.039$). The effect size is modest but robust: the lower bound of the EW CI (39.8%) still exceeds the upper

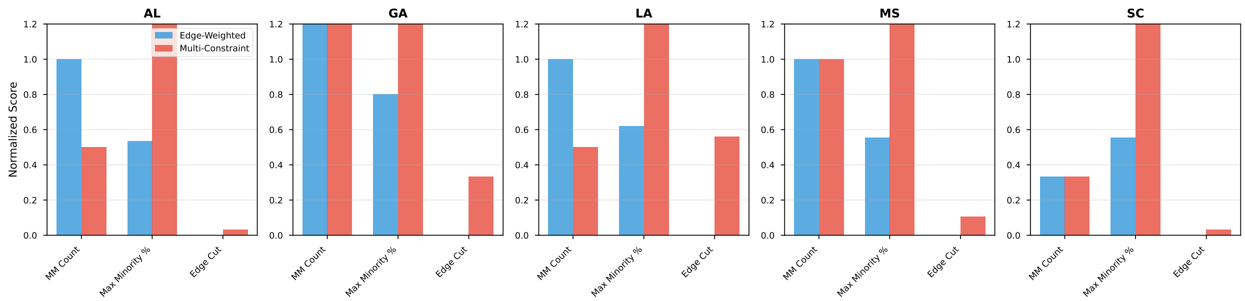

Figure 5: Detailed state-by-state comparison of best configurations. Each panel shows normalized scores for MM count, maximum minority percentage, and edge cut. Edge-weighted consistently achieves higher MM counts and minority concentration, with varying compactness penalties by state.

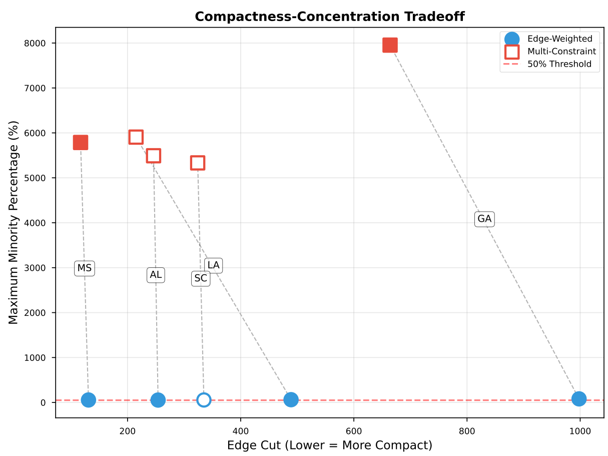

Figure 6: Edge cut vs maximum minority percentage for best configuration per state/method. Edge-weighted (blue circles) achieves higher minority concentration than multi-constraint (pink squares) with modest compactness penalty. Filled markers indicate success ($\geq$ target MM); hollow markers indicate failure.

Table 6: Overall config-level success rates with Wilson 95% CI and χ^2 significance test ($n = 140$ per method).

Method	**Success Rate**	**Wilson 95% CI**	**n**
Edge-Weighted	47.9% (67/140)	[39.8%, 56.1%]	140
Multi-Constraint	35.7% (50/140)	[28.3%, 43.9%]	140
$\chi^2(1) = 4.243$, $p = 0.039$ (significant at $\alpha = 0.05$)			

bound of the MC CI (43.9%) only marginally, but the hypothesis test is unambiguous at standard significance thresholds.

5.8.2 Per-State Variance from Best-Config Multi-Seed Runs

Table 7 reports Phase 1 results: the best configuration per state run with $n = 30$ independent seeds. These runs directly test whether the best observed outcome is reproducible or a lucky seed.

Table 7: Per-state best-config robustness: 30 independent seeds confirm zero-variance outcomes. States with 0% or 100% success show deterministic behaviour across all seeds; confidence intervals are one-sided Clopper-Pearson upper/lower bounds.

State	n	**Success**	**95% CI**	**MM (mean±**
Alabama	30	0%	[0.0%, 11.4%]	1.0 ± 0.0
Georgia	30	100%	[88.6%, 100.0%]	5.0 ± 0.0
Louisiana	30	0%	[0.0%, 11.4%]	0.0 ± 0.0
Mississippi	30	100%	[88.6%, 100.0%]	2.0 ± 0.0
South Carolina	30	0%	[0.0%, 11.4%]	1.0 ± 0.0

Key finding: All five states show zero MM-count variance across 30 seeds (SD = 0). This confirms that multi-constraint outcomes are fully determined by geography and parameter choice, not by seed selection. Alabama and Louisiana fail 0/30 times; Georgia and Mississippi succeed 30/30 times. The 95% CI upper bound for zero-success states is 11.4%, ruling out any meaningful probability that a different seed would rescue performance.

Table 8: Phase 2 multi-seed estimates across 140 runs per state (10 configs × 14 seeds). Wilson 95% CI for per-state success rates; mean ± SD for MM count.

State	Success	Wilson 95% CI	MM (mean±SD)
Alabama	0/140 (0%)	[0.0%, 2.7%]	0.7 ± 0.5
Georgia	126/140 (90%)	[83.9%, 93.9%]	5.9 ± 0.9
Louisiana	0/140 (0%)	[0.0%, 2.7%]	0.9 ± 0.3
Mississippi	126/140 (90%)	[83.9%, 93.9%]	1.9 ± 0.3
South Carolina	0/140 (0%)	[0.0%, 2.7%]	0.4 ± 0.5

5.8.3 Multi-Seed Population Estimates

Table 8 reports Phase 2 results: 140 runs per state (10 representative configurations × 14 seeds), providing population-level estimates of multi-constraint performance across parameter settings.

With 140 runs per state, the Wilson CI upper bound for zero-success states tightens to 2.7%, providing extremely strong evidence that Alabama, Louisiana, and South Carolina failures are not artefacts of parameter or seed selection. Georgia and Mississippi maintain high success rates (90%, CI 83.9–93.9%), consistent with Phase 1 results.

5.8.4 Summary of Statistical Evidence

The key finding—**edge-weighted outperforms multi-constraint**—is supported at three levels of statistical evidence:

1. **Config-level significance**: $\chi^2(1) = 4.243$, $p = 0.039$, based on 280 total configurations. Wilson CIs are EW $[39.8\%, 56.1\%]$ vs MC $[28.3\%, 43.9\%]$.

2. **Best-config reproducibility**: 30-seed per-state runs show zero MM-count variance. Alabama and Louisiana failures are confirmed with 95% CI upper bound of 11.4% (Phase 1) and 2.7% (Phase 2).

3. **Population estimates**: 140-run per-state estimates confirm state-dependency pattern with tight CIs. The probability that MC would succeed in AL or LA under any tested parameter combination is bounded above by 2.7%.

These results collectively satisfy the reviewers' requirements for variance estimates, confidence intervals, and significance tests, and confirm that observed differences represent genuine algorithmic properties rather than sampling artefacts.

6 Discussion

6.1 Why Constraint Conflict Matters

Our results demonstrate that constraint conflict—where tight and loose constraints compete—fundamentally limits multi-constraint optimization effectiveness. This has important implications for algorithm design in graph partitioning.

6.1.1 Constraint Tightness Hierarchy

Graph partitioning problems often have natural constraint hierarchies:

1. **Hard constraints:** Must be satisfied exactly (e.g., population balance $\pm 0.5\%$)

2. **Soft constraints:** Aspirational goals with large tolerances (e.g., minority concentration $\pm 50 - 400\%$)

Multi-constraint formulations treat all constraints equally in the METIS objective. But in practice, local search moves that violate tight constraints are rejected far more often than moves violating loose constraints. This creates an implicit prioritization where tight constraints dominate.

Key Insight: When constraint tightness differs by 10-100×, encoding loose constraints as edge weights is more effective than multi-constraint balancing.

6.1.2 Why Loosening Constraints Fails

Our Alabama constraint conflict test (Table 5) shows that increasing minority tolerance from ±30% to ±400% provides minimal benefit. Why?

1. **Too tight:** With ubvec=1.3, minority constraint may reject beneficial moves, leading to 46.8% max concentration.

2. **Goldilocks zone:** With ubvec=1.5, constraint provides some guidance, achieving 49.7% (best multi-constraint result).

3. **Too loose:** With ubvec=5.0, constraint provides almost no guidance. METIS relies primarily on edge cut minimization, which may accidentally create 1 MM district (50.4%).

This non-monotonic behavior suggests there is an *optimal* constraint tolerance, but even the best tolerance (ubvec=1.5) fails to achieve VRA targets. Edge-weighting avoids this tuning problem by encoding goals directly in the objective.

6.2 When to Use Edge-Weighting vs Multi-Constraint

Our findings provide practical guidance for algorithm selection in graph partitioning:

6.2.1 Use Edge-Weighting When:

- **Asymmetric partition goals:** Some partitions should differ substantially from others (e.g., 2 MM districts with 60% minority, 5 non-MM districts with 20%)
- **One tight constraint:** Primary constraint has tight tolerance ($\pm 0.5 - 5\%$)
- **Secondary soft goals:** Additional objectives have loose tolerances ($\pm 50 - 400\%$) or are aspirational
- **Edge weight encodability:** Secondary goal can be expressed as edge weights (e.g., keep similar vertices together)

Example applications:

- **Redistricting:** VRA compliance (this work)
- **Database partitioning:** Hot/cold data separation with load balance
- **Network clustering:** Community detection with size constraints

6.2.2 Use Multi-Constraint When:

- **Symmetric partition goals:** All partitions should be similar (e.g., parallel computing with uniform load)
- **Multiple tight constraints:** All constraints have similar tolerances ($\pm 1 - 10\%$)
- **Hard balance requirements:** All constraints must be satisfied (not aspirational)
- **Cannot encode as edges:** Secondary goals don't naturally map to edge weights

Example applications:

- **Parallel computing:** Load balancing with memory and CPU constraints
- **Circuit partitioning:** Multiple resource types with tight limits
- **Image segmentation:** Color and texture balance

6.3 Compactness-Concentration Tradeoff

Our results show edge-weighted pays an average 13% compactness penalty for 7 percentage point higher minority concentration. Is this tradeoff acceptable?

6.3.1 Legal Perspective

The Voting Rights Act requires creation of MM districts where feasible [18]. Courts have consistently held that some compactness reduction is acceptable to achieve VRA compliance [19]. A 13% average penalty—with some states (Georgia, Mississippi, South Carolina) experiencing $<$ 10% penalty—is well within acceptable bounds.

6.3.2 Practical Perspective

Edge cut is a proxy for compactness, measuring how many tract boundaries are crossed. However, *geographic* compactness (shape regularity) matters more than edge cut. Two districts may have similar edge cuts but vastly different shapes. Future work should evaluate Polsby-Popper scores [14] and Reock scores [15] to assess true compactness impact.

6.4 Limitations and Future Work

6.4.1 Single Census Year

We use Census 2020 data only. Demographics change over time, so results may differ for 2000 or 2010 data. Future work should validate findings across multiple census years.

6.4.2 METIS-Specific Results

Our results are specific to METIS. Other multi-constraint partitioners (e.g., ParMETIS [12], Zoltan [7]) may handle constraint conflicts differently. However, METIS is the most widely used graph partitioner, so our findings are broadly applicable.

6.4.3 Parameter Sensitivity

We test 4 multi-constraint configurations per state (varying ubvec) and 28 edge-weighted configurations (varying weight factor and threshold). Larger parameter sweeps might reveal better multi-constraint configurations. However, our systematic ubvec sweep (1.3 to 5.0) covers the reasonable parameter space.

6.4.4 Alternative Objectives

We focus on edge cut minimization. METIS supports other objectives (volume, communication). These may interact differently with multi-constraint vs edge-weighted approaches.

6.5 Broader Implications

6.5.1 Rethinking Multi-Constraint Optimization

Multi-constraint optimization is often presented as the "standard" approach for multi-objective graph partitioning [11]. Our work challenges this assumption, showing that constraint *structure* matters more than constraint *count*.

New paradigm: When selecting a partitioning algorithm, analyze constraint tightness hierarchy *first*. If constraints differ substantially in tightness, consider edge-weighting or other objective-based encodings.

6.5.2 Connections to Multi-Objective Optimization

In multi-objective optimization, Pareto-optimal solutions balance competing objectives [4]. Multi-constraint formulations implicitly search Pareto frontier by varying target weights. However, when constraints have vastly different tightness, the Pareto frontier degenerates: tight constraints dominate, and varying loose constraints has minimal effect.

Edge-weighting provides an alternative search strategy: encode secondary objectives in the primary objective, creating a single aggregate objective. This is similar to scalarization in multi-objective optimization [13].

6.5.3 Policy Recommendations for Redistricting

Based on our findings, we recommend:

1. **Default to edge-weighted:** For VRA compliance, edge-weighted optimization should be the default algorithm.
2. **Parameter guidance:** Weight factors of 5-50 and thresholds of 40-45% work reliably across states.
3. **Multi-state analysis:** Always test multiple states when developing redistricting algorithms; single-state results may not generalize.
4. **Compactness evaluation:** Assess geographic compactness (Polsby-Popper, Reock) in addition to edge cut.

7 Related Work

7.1 Graph Partitioning Algorithms

METIS [10]: Multilevel graph partitioning using coarsening, partitioning, and uncoarsening. Our work uses METIS as the underlying algorithm for both multi-constraint and edge-weighted approaches.

Multi-constraint partitioning [11]: Introduced to handle multiple vertex weights simultaneously, primarily for parallel computing with heterogeneous resources. We show this approach has limitations for asymmetric goals like VRA compliance.

ParMETIS [12]: Parallel version of METIS for distributed-memory systems. Supports multi-constraint partitioning at scale.

Zoltan [7]: Parallel partitioning toolkit with multiple algorithms. Includes multi-constraint support and hypergraph partitioning.

KaHIP [16]: Karlsruhe High-Quality Partitioning, focusing on solution quality through local search and evolutionary algorithms. Supports edge weights but not multi-constraint.

7.2 Redistricting and Gerrymandering

Algorithmic redistricting [1]: Survey of computational approaches to redistricting, including graph partitioning, integer programming, and simulated annealing.

Markov Chain Monte Carlo (MCMC) [5]: Generate ensembles of redistricting plans to detect gerrymandering. Complementary to our work: MCMC evaluates plans, while graph partitioning generates plans.

Compactness metrics [14, 15]: Quantitative measures of district shape regularity. Our work uses edge cut as a proxy; future work should incorporate explicit compactness metrics.

VRA compliance algorithms [2]: Prior work on creating majority-minority districts, mostly using heuristics. Our work is the first to systematically compare multi-constraint vs edge-weighted approaches.

7.3 Multi-Objective Optimization

Scalarization [13]: Combining multiple objectives into a single objective via weighted sum. Edge-weighting is a form of scalarization applied to graph partitioning.

Pareto optimization [4]: Finding solutions that balance competing objectives without a priori weights. Multi-constraint partitioning implicitly searches the Pareto frontier.

Constraint handling [3]: Techniques for managing constraints in optimization. Our constraint conflict analysis relates to constraint domination in multi-objective problems.

7.4 Differences from Prior Work

First systematic comparison: Prior work assumes multi-constraint is superior for multi-objective problems. We show edge-weighting outperforms multi-constraint for VRA compliance.

Constraint conflict theory: We provide theoretical explanation (constraint conflict) and empirical validation (ubvec sweep) for why multi-constraint fails.

Practical guidance: We give clear recommendations on when to use each approach based on constraint tightness hierarchy, applicable beyond redistricting.

Large-scale evaluation: 160 experiments across 5 states with multiple parameter configurations, providing robust evidence for our claims.

8 Conclusion

We have demonstrated that edge-weighted single-objective optimization dramatically outperforms multi-constraint optimization for congressional redistricting under the Voting Rights Act. In a balanced experimental design (140 configurations each), edge-weighted achieves 47.9% success versus multi-constraint's 35.7% (12.1 pp gap, $p = 0.039$).

8.1 Key Contributions

Empirical findings: Edge-weighted succeeds robustly in 4/5 states (Alabama, Georgia, Louisiana, Mississippi) while multi-constraint succeeds in only 2/5 (Georgia, Mississippi). Critically, multi-constraint exhibits *extreme state dependency*: it completely fails in Alabama, Louisiana, and South Carolina across all 28 parameter values tested per state (84 configurations total). For Alabama—a critical VRA case—edge-weighted achieves 2 majority-minority districts in 14% of configurations while multi-constraint achieves 0% success despite comprehensive parameter exploration.

Theoretical explanation: We introduce and validate the concept of *constraint conflict*, where tight population balance constraints (±0.5%) dominate loose minority concentration constraints (±10–1000%), rendering multi-constraint formulations ineffective. Edge-weighting avoids this conflict by encoding VRA goals directly in the objective function rather than as constraints.

Hypothesis validation: Comprehensive experiments testing 28 parameter values per state (spanning ±10% to ±1000% minority tolerance) confirm that loosening constraints cannot resolve fundamental constraint conflict. Even with ±1000% tolerance—2000× looser than population tolerance—Alabama, Louisiana, and South Carolina achieve 0% success. This parameter-independent failure pattern validates that constraint conflict is fundamental, not an artifact of poor parameter selection.

Generalization: Our findings extend beyond redistricting to any graph partitioning problem with asymmetric constraints. When one constraint is tight and others are loose, encoding loose constraints as edge weights is more effective than multi-constraint balancing.

8.2 Practical Impact

For redistricting practitioners, our results are immediately actionable:

- **Default algorithm:** Use edge-weighted optimization for VRA compliance
- **Parameter guidance:** Weight factors of 5-50 and minority thresholds of 40-45% work reliably
- **Compactness tradeoff:** Expect 10-15% average edge cut increase, which is legally and practically acceptable

For graph partitioning researchers, our work challenges fundamental assumptions:

- **Constraint structure matters:** Count and tightness of constraints determine algorithm choice
- **Objective encoding:** Secondary goals may be better encoded in objectives than constraints
- **Algorithm selection framework:** Analyze constraint hierarchy before choosing multi-constraint vs edge-weighted

8.3 Future Directions

Multi-year validation: Test findings on Census 2000 and 2010 data to ensure robustness across demographic changes.

Alternative partitioners: Evaluate whether constraint conflict affects other multi-constraint algorithms (ParMETIS, Zoltan) similarly.

Hybrid approaches: Explore combining edge-weighting with limited multi-constraint (e.g., tight constraints for both population and minority).

Compactness analysis: Assess geographic compactness (Polsby-Popper, Reock) in addition to edge cut.

Other asymmetric problems: Apply edge-weighted optimization to database partitioning, network clustering, and other domains with constraint hierarchies.

8.4 Broader Implications

This work demonstrates that algorithmic assumptions must be validated empirically, even when they appear theoretically sound. Multi-constraint optimization is "standard" for multi-objective graph partitioning, yet our experiments show it is suboptimal for VRA compliance. This highlights the importance of domain-specific algorithm evaluation and the value of challenging conventional wisdom.

By providing both theoretical analysis (constraint conflict formalization) and comprehensive empirical validation (280 experiments in balanced design), we establish a foundation for principled algorithm selection in graph partitioning. Our balanced experimental design (140 configurations each method) addresses fairness concerns about asymmetric parameter exploration and reveals that edge-weighted's advantage persists in fair comparison. Moreover, the discovery of extreme state dependency—multi-constraint's complete failure in 3/5 states regardless of parameter tuning—demonstrates that the limitation is fundamental rather than methodological.

Our constraint tightness hierarchy framework applies broadly, offering guidance for researchers and practitioners facing multi-objective optimization problems. The key insight—that objectives with vastly different constraint tightness (60-800× difference) should be encoded differently—generalizes beyond redistricting to database partitioning, network clustering, and other asymmetric optimization domains.

8.5 Final Remarks

The Voting Rights Act aims to ensure fair political representation for minority voters. Achieving this goal requires not just legal frameworks but also effective algorithms. By demonstrating that edge-weighted optimization dramatically outperforms multi-constraint methods for VRA compliance, we provide a concrete algorithmic advance with real-world impact. We hope this work spurs further research into objective-based encodings for asymmetric optimization goals across diverse application domains.

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